Bhavya Shah

bhavya.mehul.shah@gmail.com • +91 7016017279 • bhavyaresume.vercel.app
LinkedIn

EDUCATION

Nirma University, Ahmedabad 2023–2027
B.Tech in Computer Science and Engineering CGPA: 8.74

SKILLS

Technical Skills: C/C++, JavaScript, Node.js, Next.js, Python3, MySQL
Coursework: Deep Learning, Data Communication, Operating Systems, Database Management System, Object-Oriented Programming, Computer Networks, Computer Architecture, Electronic Systems and Engineering, Microcontroller and Interfacing, Cloud Computing

KEY COMPETENCIES & LEADERSHIP

A proactive and organized communicator with strong critical thinking skills and a passion for technology, research, and teaching.

- Technical Director, ACES – Lead at organizing various technical events such as Insignia Hackathon, Summer Internship Talks, and many more
- Winner, DotSlash 9.0 (SVNIT, Surat)
- 2nd Runner Up, Hack-a-Mind (Nirma University, SUNY Binghamton University)
- Finalist at FinShield Hackathon 2025 (IIT Hyderabad)

PROJECTS

RiskAnalyzer: Alternative Data Credit Risk Model
Developed a machine learning model and full-stack web application to predict credit default probability using non-traditional data sources (e.g., mobile usage, utility payments). Built with *Python, Flask, and React*, enabling inclusive risk assessments for 'thin-file' customers.
Live Demo: defaultprediction-frontend.onrender.com
GitHub: github.com/GlyphCoder/RiskAnalyzer

VoxelWorld
Lightweight, browser-based 3D voxel sandbox featuring procedural environments, dynamic time-of-day lighting, and a custom built-in Neural Network for generating instant architectural structures.
Live Demo: voxelworld-builder.vercel.app
GitHub: github.com/GlyphCoder/voxelworld

StoryArc AI : Episodic Intelligence Engine
An AI-powered episodic intelligence engine that transforms your story ideas into fully analyzed, production-ready vertical series (90-second episodes). It leverages *Google Gemini AI* to provide comprehensive narrative analysis and creative guidance.
GitHub: github.com/GlyphCoder/StoryArc-AI

RESEARCH WORK

TinyML-Based Trojan Detection Framework in Smart Grid Systems *Accepted for publication at [The 15th International Smart Grids Conference], [2025]*
Proposed a TinyML-driven intrusion detection model for smart grids achieving 98.39% accuracy with 87% storage compression, demonstrating efficient, real-time, edge-based cybersecurity.

Lean Memory, Deep Mastery: Toward Thoughtful Dialogue through Continual Reinforcement in LLM Tutors *Completed, Under Review*

Introduced a continual reinforcement learning framework with cognitive feedback to transform LLMs into reflective, adaptive tutors that learn from dialogic interactions.

Machine and Reinforcement Learning-Based Collaborative Framework for Drone Navigation in Battlefield Environment *Preprint*

Developed a Q-learning–based reinforcement learning framework integrated with ML classifiers for autonomous UAV navigation, achieving enhanced adaptability and collision avoidance in dynamic battlefield environments.

TinyML-Based Efficient Framework for Industrial IoT Communication Using 5G Network Slicing *Completed, Under Review*

Designed a scalable 5G network slicing system with TinyML quantized models (LinearSVM, DecisionTree, ANN), improving latency and efficiency while maintaining 82.1% accuracy with 77% smaller model size.